

Cooperative Caching of Multi-view 3D Video in LEO Satellite Networks with Dynamic View Selection and Inter-Satellite Links

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Outline

- ▶ Introduction
- ▶ System Models
- ▶ Problem Formulation
- ▶ Proposed Algorithm
- ▶ Simulation Results
- ▶ Conclusion

Introduction

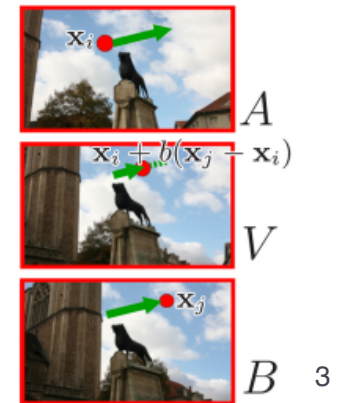
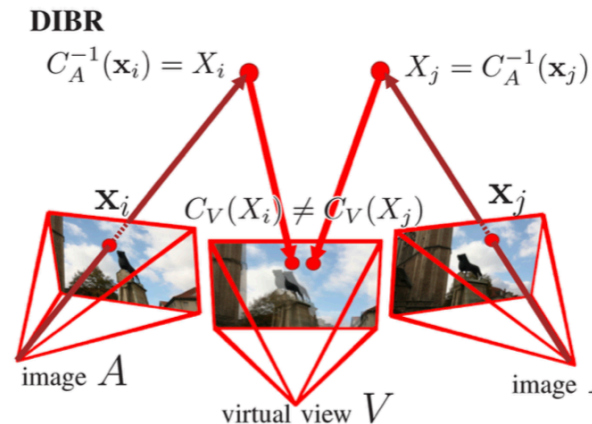
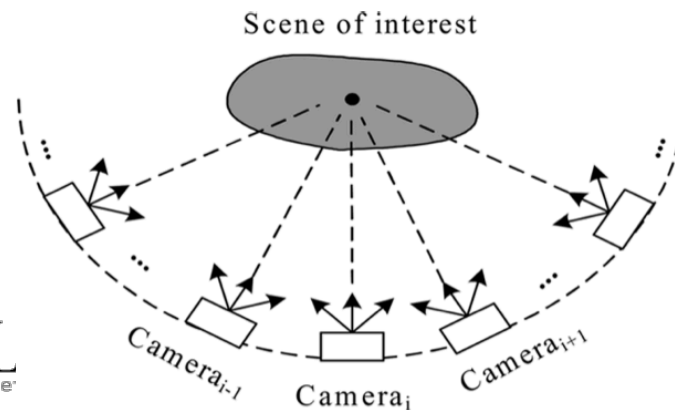
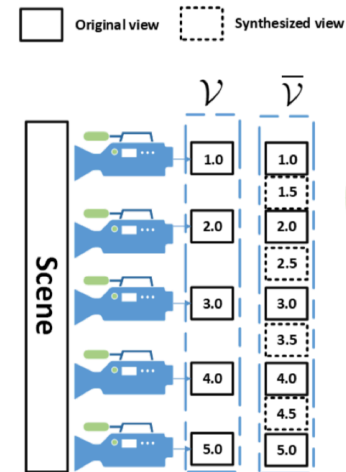
- ▶ Cooperative caching for **multi-view 3D videos** in **LEO** satellite networks that optimizes content placement and delivery by utilizing **inter-satellite links (ISL)** and **DIBR synthesis**

- ▶ **Multi-view 3D Video (MVV)**

- ▶ Enables users to interactively select preferred views from different directions
- ▶ Require larger storage and bandwidth consumption compared to single-view video

- ▶ **Depth-Image Based Rendering Synthesis (DIBR)**

- ▶ Synthesis by left and right reference views, less views are required
- ▶ Cost incurred by DIBR is sufficiently small for modern devices



[8] Q. Chen, G. Giambene, L. Yang, C. Fan and X. Chen, "Analysis of Inter-Satellite Link Paths for LEO Mega-Constellation Networks," in *IEEE Transactions on Vehicular Technology*, vol. 70, no. 3, pp. 2743-2755, March 2021

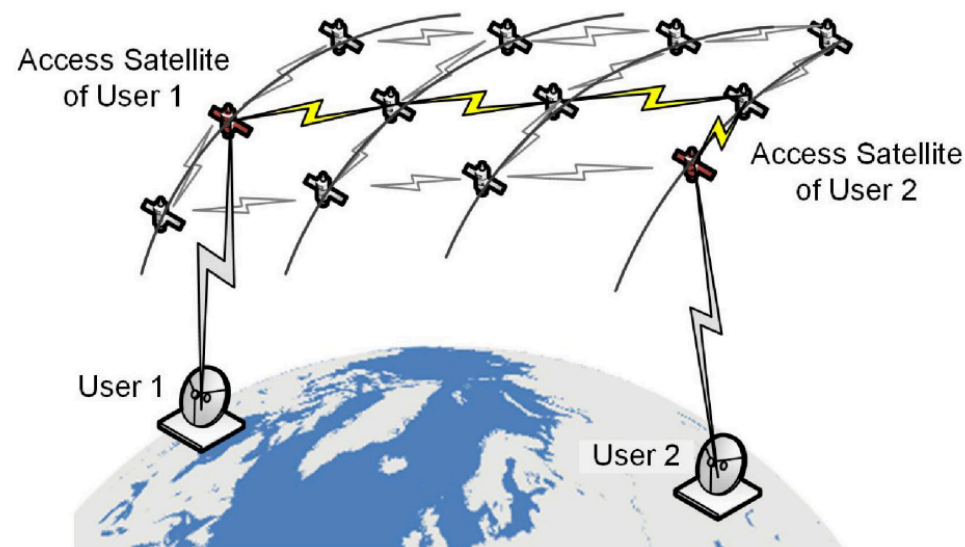
[10] R. Zhao, Y. Ran, J. Luo and S. Chen, "Towards Coverage-Aware Cooperative Video Caching in LEO Satellite Networks," *GLOBECOM 2022 - 2022 IEEE Global Communications Conference*, Rio de Janeiro, Brazil, 2022

[17] Ruili Zhao, Jiangtao Luo, and Yongyi Ran. 2023. Coverage-Aware Cooperative Caching and Efficient Content Distribution Schemes in LEO Satellite Networks. In *Proceedings of the 1st ACM MobiCom Workshop on Satellite Networking and Computing (SatCom '23)*. Association for Computing Machinery, New York, NY, USA, 31–36.

Introduction

► Cooperative caching on LEO satellite network

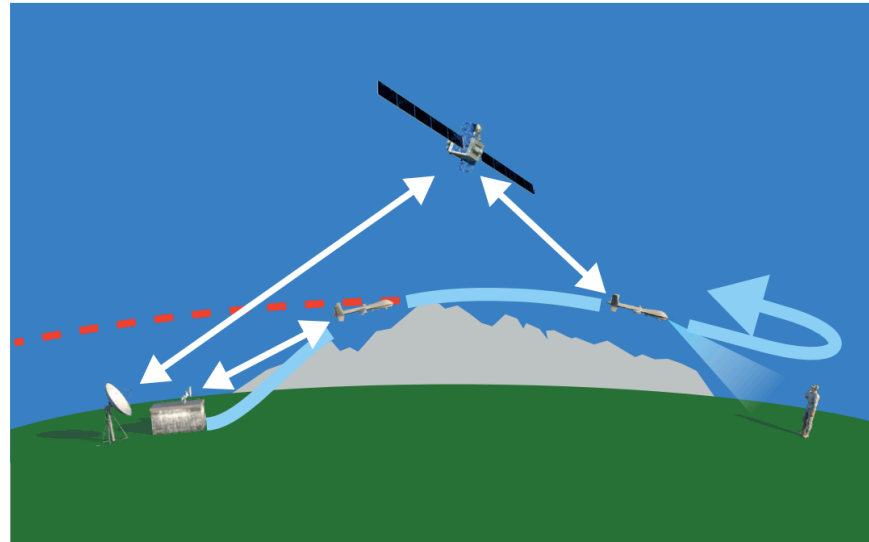
- Wide coverage for global users
- Caching view on satellites shorten service delay and speeds up view sharing
- Pre-caching on LEO to reduce repetitive view traffic and service delay
- **ISL** enable cooperative content sharing and extend effective cache coverage across the satellite [8]
- General content and 2D video for single-view has been studied to reduce delay [10,17]



Introduction

▶ Video Delivery with LEO Satellites

- ▶ Honeywell UAV Satcom
- ▶ Airbus and OneWeb
- ▶ EarthNow
- ▶ Panasonic Avionics



1. Honeywell UAV Satcom https://www.youtube.com/watch?v=5tjDFG9gXIY&ab_channel=HoneywellAerospaceTechnologies
2. LEO satellite communications services in development with Airbus and OneWeb <https://militaryembedded.com/comms/satellites/leo-satellite-communications-services-in-development-with-airbus-and-oneweb>
3. Earth Now – Climate Change: Vital Signs of the Planet - NASA <https://climate.nasa.gov/earth-now/>
4. Panasonic Avionics-What is Low Earth Orbit (LEO) and How Does it Benefit Airline Connectivity? <https://www.panasonic.aero/blog/blog-post/what-is-low-earth-orbit-leo-and-how-does-it-benefit-airline-connectivity>

- [13] Leng, Tao, Yuanyuan Xu, Gaofeng Cui, and Weidong Wang. 2021. "Caching-Aware Intelligent Handover Strategy for LEO Satellite Networks" *Remote Sensing* 13
- [22] S. Bhandari, T. X. Vu and S. Chatzinotas, "User-Centric Flexible Resource Management Framework for LEO Satellites With Fully Regenerative Payload," in *IEEE Journal on Selected Areas in Communications*, vol. 42, no. 5, pp. 1246-1261, May 2024
- [23] J. Tang *et al.*, "Cooperative Caching in Satellite-Terrestrial Integrated Networks: A Region Features Aware Approach," in *IEEE Transactions on Vehicular Technology*, vol. 73, no. 7, pp. 10602-10616, July 2024

Related Works

	Objective	Content Type	Network Type	DIBR Synthesis	Utilizing ISL	Coverage Scope	Methodology
[9]	Minimize total cost	Multi-view video	Mobile proxy	O	X	Urban only	MDP + online algorithm
[10]	Minimize average delay	2D video	LEO satellite	X	O	Global	MADDPG
[11]	Minimize total delay	General files	LEO Satellite	X	X	Global	LDD-Based Algorithm
[12]	Minimize power consumption	General files	Single LEO	X	X	Regional Cluster	CCP-based Algorithm
[13]	Reduce handover failure rate	General files	LEO satellite	X	O	Global	DRL
[22]	Minimize average delay	General files	LEO satellite	X	O	Global	K-means + SCA + DL-CSI
[23]	Minimize content delivery delay	General files	LEO satellite	X	O	Global	Region-aware (game-theoretic)
This work	Minimize total cost	Multi-view video	LEO satellite	O	O	Global	Online algorithm

- [9] M. Yeh, C. -H. Wang, D. -N. Yang, J. -T. Lee and W. Liao, "Mobile Proxy Caching for Multi-View 3D Videos With Adaptive View Selection," in *IEEE Transactions on Mobile Computing*, vol. 21, no. 8, pp. 2909-2921, 1 Aug. 2022
- [10] R. Zhao, Y. Ran, J. Luo and S. Chen, "Towards Coverage-Aware Cooperative Video Caching in LEO Satellite Networks," *GLOBECOM 2022 - 2022 IEEE Global Communications Conference*, Rio de Janeiro, Brazil, 2022
- [11] Y. Hao, Z. Song, Z. Zheng, Q. Zhang and Z. Miao, "Joint Communication, Computing, and Caching Resource Allocation in LEO Satellite MEC Networks," in *IEEE Access*, vol. 11, pp. 6708-6716, 2023
- [12] D. Han, H. Peng, H. Wu, W. Liao and X. S. Shen, "Joint Cache Placement and Content Delivery in Satellite-Terrestrial Integrated C-RANs," *ICC 2021 - IEEE International Conference on Communications*, Montreal, QC, Canada, 2021



Motivation and Research Challenges

- ▶ Terrestrial network to LEO satellite migration
 - ▶ Limit computing resources and storage
 - ▶ Unstable communication links
 - ▶ Intermittent ground station connectivity
 - ▶ Each satellite performs independently and manages its cache
- ▶ Adding DIBR synthesis in the LEO satellite network
 - ▶ Multi-option decision making becomes complex
- ▶ **New challenges:**
 - ▶ Cooperative **LEO MVV caching** with **ISL** and **DIBR** synthesis
 - ▶ The balance between **global view popularity** and **DIBR possibility**
 - ▶ Trade-off between diverse **user request ranges** and **cache utilization** efficiency

Contribution

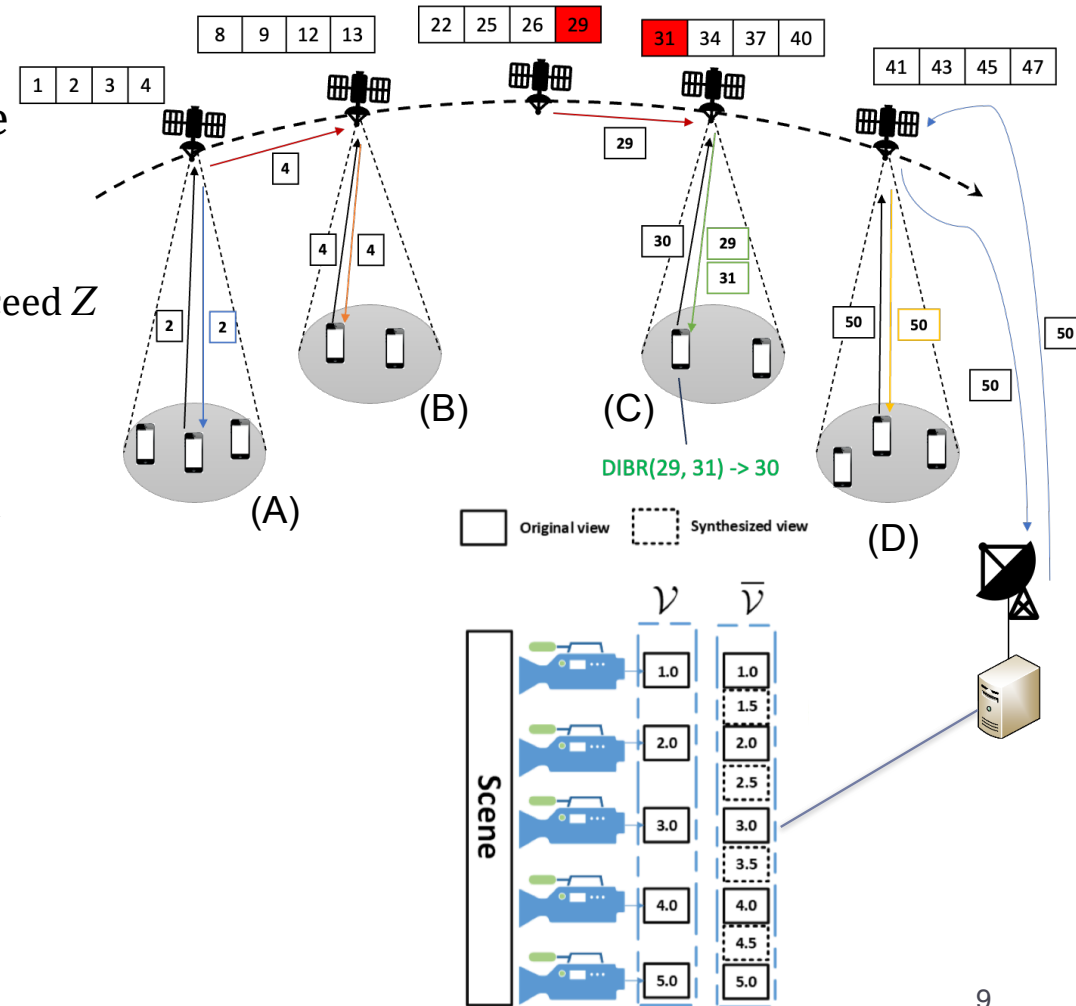
- ▶ We make the first attempt to investigate LEO satellite networks, MVV, and DIBR synthesis for cooperative caching, formulating the **Multi-view Cooperative Caching Management with Synthesis** (MCCMS) problem
- ▶ We prove MCCMS is **NP-hard** and propose the AMSCC algorithm with **competitive ratio** and **time complexity** analysis
- ▶ Simulations show AMSCC outperforms baselines by 15-35% in cache **hit rate** and 10-30% in system **cost reduction**



System Model: Network Architecture

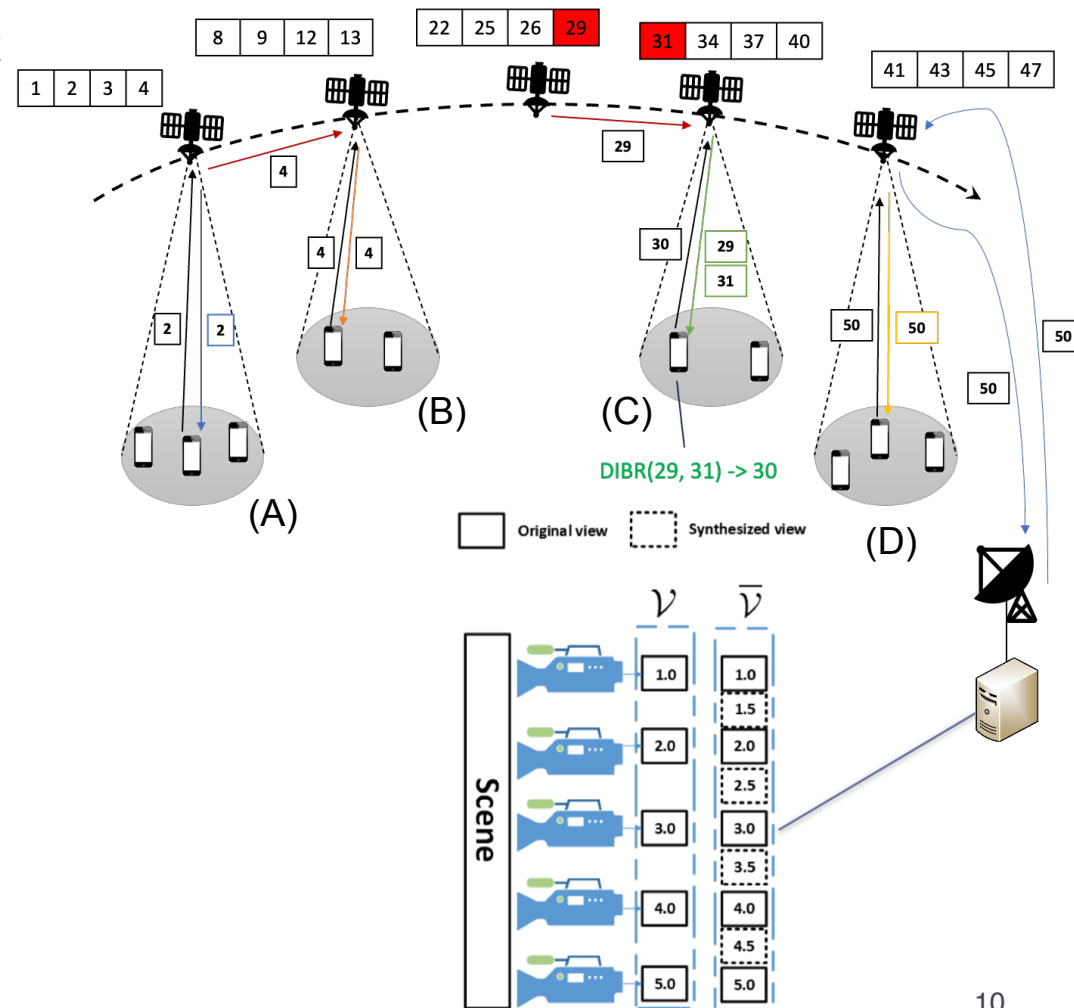
- ▶ **Users:** mobile user terminals requesting MVV
- ▶ **LEO satellites:** constellation with limited cache storage
 - ▶ Every LEO satellite has the same cache storage
 - ▶ The total size of cached views in each LEO cannot exceed Z

$$\sum_{i \in V_n(t)} z_i \leq Z$$
- ▶ **Ground stations:** serve as the primary content source in the cache system
- ▶ **MVV content:** videos with multiple camera viewpoints
 - ▶ V : the universal set of views in a MVV
 - ▶ $V_n(t)$: the set of views stored in LEO n at time t
 - ▶ z_i : the size of view i



System Model: Cache Operation

- ▶ When LEO satellite receives user's request
- ▶ **Cache Hit**
 - A. Cached in access LEO
 - B. Neighboring LEOs could sharing requested view via *ISL*
 - C. Return left and right views to user and by synthesizing requested view (DIBR)
- ▶ **Cache Miss**
 - D. The above cases not satisfied and should fetch the request view from GS (source)



System Model: Cache Decision Variables

► **These are the decision variables that define different view sets for LEO n :**

- $V_n^{syn}(t)$: views are **not cached** in local LEO n or neighbors but **could be synthesized by DIBR** constraint D
 - $V_n(t) \cap V_{n\pm 1}(t) \cap V_n^{syn}(t) = \emptyset$
- $V_n^f(t)$: views be **fetches from GS**
- $V_n^{DIBR}(t)$: views are **synthesized on the user device** using **two reference views**
- $V_n^{ISL}(t)$: views **retrieved from adjacent satellites** utilizing **ISL**
- $V_n^s(t)$: includes all views that LEO n **sends to users**, which must fall within the request range
- $V_n^e(t)$: views that would be **removed** when the satellite's **cache is full**

System Model: Satellite Model [14][15]

- ▶ Service Area Model: $A_n(t) = \{x(R, \theta, \Phi) \mid \sin \theta \sin \theta_n \cos(\Phi - \Phi_n) + \cos \theta \cos \theta_n \leq \cos \psi\}$

- ▶ R : the radius of earth

- ▶ Φ_n : the longitude of satellite n

- ▶ $\theta_n = \left| \frac{\pi}{2} - n_{latitude} \right|$

- ▶ ψ : **half cone angle** between covered area and Earth's core

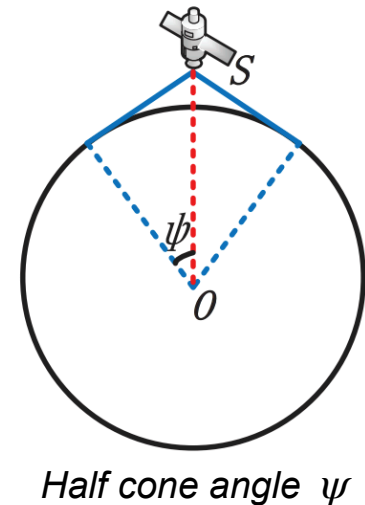
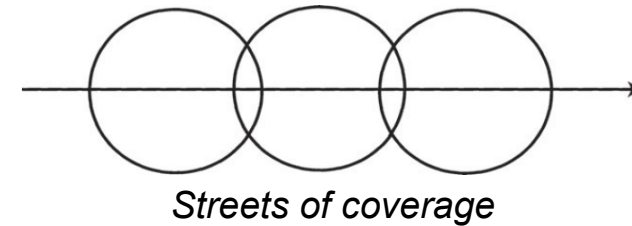
- ▶ The point set of all points in the area covered by satellites: $A_{SOC}(t) = \bigcup_{n=1}^n A_n(t)$

- ▶ Coverage Model:

- ▶ $\psi = \arccos\left(\frac{R}{R+h} \cos \theta_{min}\right) - \theta_{min}$

- ▶ The coverage area $S = 2\pi R^2(1 - \cos \psi)$

- ▶ A larger value of ψ implies the decrease in the coverage



System Model: Communication Model [16][17]

- ▶ LEO_n to $User_k$ transmission:

- ▶ $R_{n,k} = B_{n,k} \log_2 \left(1 + \frac{P_n \cdot g_{n,k}}{\sigma^2} \right)$

- ▶ Inter-satellite link transmission:

- ▶ $R_{n,n+1}^{ISL} = B^{ISL} \log_2 \left(1 + \frac{P_{ISL} \cdot g_{n,n+1}}{\sigma^2} \right)$

- ▶ GS_g to LEO_n transmission:

- ▶ $R_{g,n} = B^{GS} \log_2 \left(1 + \frac{P_{GS} \cdot p_{g,n}}{\sigma^2} \right)$

- ▶ Bandwidth constraint

- ▶ $\sum_{n \in \mathcal{N}} B_n(t) \leq B_n^{max}$

- ▶ B : the channel bandwidth

- ▶ P : the transmit power

- ▶ G : the channel gains

- ▶ $d_{n,k} = -R \sin \theta_{n,k} + \sqrt{(R \sin \theta_{n,k})^2 + h^2 + 2hR}$

- ▶ the distance between LEO_n and user k

- ▶ $\theta_{n,k}$: the elevation angle of user k with LEO_n

- ▶ $g_{n,k} = G d_{n,k}^{-\alpha}$

- ▶ G : the constant power gains factor by amplifier and antenna



System Model: View Request Model [9]

- ▶ User request: $R_n(t)$
 - ▶ A request here is a range of views for users to adaptively change the view angle at different time
 - ▶ Users can only be served when within satellite constellation coverage area $k(R, \theta, \Phi) \in A_{SOC}(t)$
- ▶ The request range of the users: $\left[R_n(t) - \lfloor \frac{B}{2} \rfloor, R_n(t) + \lfloor \frac{B}{2} \rfloor \right]$
 - ▶ B is the size of a request range
 - ▶ Fetch a range of views for each user to **support the rapid view change**
- ▶ DIBR synthesis
 - ▶ a target view v can be reconstructed from v_l (left) and v_r (right) if the $|v_l - v_r| \leq D$
 - ▶ D is a maximum synthesis distance constraint that ensures rendering quality

Problem Formulation

► Problem:

- Multi-view Cooperative Caching Management with Synthesis (MCCMS)

- An instantaneous cost function $C(t)$ at time t for each user request: [9,18,28]

$$C(t) = \underbrace{c_m}_{\text{The penalty cost for a cache miss}} \sum_i 1_{\{i \notin V_n(t) \cup V_{n+1}(t) \cup V_n^{syn}(t)\}} + \sum_{i \in V_n^f(t)} \underbrace{c_{n,f}(z_i, d_i^{GS})}_{\text{The cost to fetch view } i \text{ from GS}} + \sum_{i \in V_n^s(t)} \underbrace{c_{n,s}(z_i, d_i)}_{\text{The cost to return view } i \text{ to user}} \\ + \underbrace{c_p(v_l, v_r)}_{\text{DIBR synthesis cost}} \sum_j 1_{j \in \{V_n^{DIBR}(t)\}} + \sum_{i \in V_n^{ISL}(t)} h \cdot \underbrace{c_{ISL}(z_i, d_i^{ISL})}_{\text{The cost to transmit view } i \text{ via ISL}}$$

$$c_p(v_l, v_r) = \alpha \cdot |v_l - v_r| + T_{DIBR}$$



Problem Formulation

▶ Objective function:

- ▶ minimize the total cost $C(t)$ in T time slots for each user request

▶ Constraints:

- ▶ LEO cache storage constraint
- ▶ DIBR synthesis constraint
- ▶ Satellite coverage constraint
- ▶ Bandwidth constraint

Theorem 1. MCCMS is NP-hard

- ▶ Reduce from *Partition Problem* [24]

$$C(t) = c_m \sum_i 1_{\{i \notin V_n(t) \cup V_n^{syn}(t)\}} + \sum_{i \in V_n^f(t)} c_{n,f}(z_i, d_i^{GS}) + \sum_{i \in V_n^s(t)} c_{n,s}(z_i, d_i) \\ + c_p(v_l, v_r) \sum_j 1_{j \in \{V_n^{DIBR}(t)\}} + \sum_{i \in V_n^{ISL}(t)} h \cdot c_{ISL}(z_i, d_i^{ISL})$$

$$\min \sum_{t=1}^T C(t) \\ \text{s.t. } C_1 : \sum_{i \in V_n(t)} z_i \leq Z, \quad \forall n \in \mathcal{N}, \forall t$$

$$C_2 : |v_l - v_r| \leq D, \quad \forall v_l, v_r \text{ used for synthesis}$$

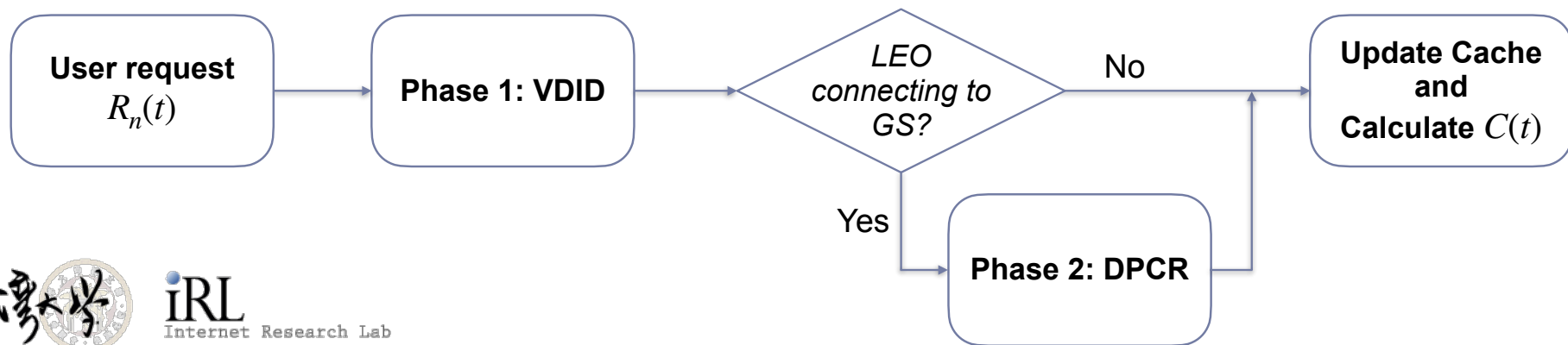
$$C_3 : k(R, \theta, \Phi) \in A_{\text{SOC}}(t), \quad \forall k \in \mathcal{K}$$

$$C_4 : \sum_{n \in \mathcal{N}} B_n(t) \leq B_n^{\max}$$



Proposed Algorithm

- ▶ Adaptive Multi-View Satellite Cooperative Caching (AMSCC) structure:
 - ▶ **Phase 1: View Delivery with ISL and DIBR (VDID)**
 - ▶ Online optimization approach due to dynamic requests and satellites moving
 - ▶ *Cooperative Delivery Cost (CDC)*
 - Determines the optimal view sets to minimize costs $C(t)$ while satisfying user range requests
 - ▶ **Phase 2: DIBR and Popularity-aware Cache Replacement (DPCR)**
 - ▶ *Popularity-Synthesis Score (PSS)*
 - Manages cache replacement during GS contacts by balancing global view popularity with synthesis potential



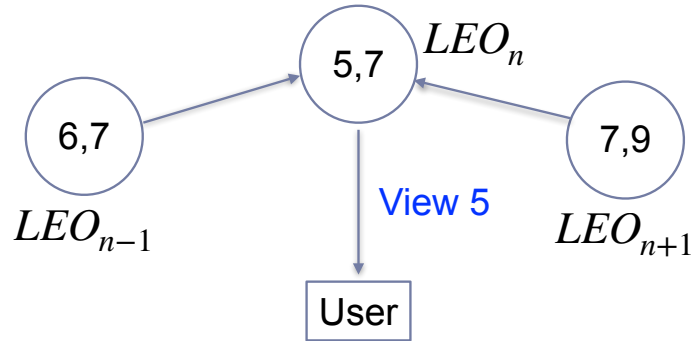
Phase 1: View Delivery with ISL and DIBR (VDID)

- ▶ For each user request (a range of views $[h, l]$)
 - ▶ Checks for cache hit from **local** or **neighboring** LEO, otherwise fetches from **GS**
- ▶ Cooperative Delivery Cost (CDC)
 - ▶ Minimize instantaneous cost of serving each user request
 - ▶ $\mu_{h,j} = \min_{\max\{j-D, h\} \leq i < j} \{ \tau_j + \mu_{h,i} + [\alpha(j-i) + T_{\text{DIBR}}](j-i-1) \}$ is the **minimum cost** of view set $[h, j]$
 - ▶ τ_j is the total cost for view j
 - **Local cache hit**: $\tau_j = c_{n,s}$ (transmission cost)
 - **ISL retrieval**: $\tau_j = c_{\text{ISL}} + c_{n,s}$ (ISL cost + transmission cost)
 - **GS fetch**: $\tau_j = c_m + c_{n,f} + h \cdot c_{\text{ISL}} + c_{n,s}$ (miss cost + fetch cost + ISL cost + transmission cost)
 - ▶ Uses **DIBR** if references are present to avoid cache miss
- ▶ Determine the different view sets
- ▶ Evict views with lowest popularity (following Zipf distribution [25]) if cache is full

Phase 1: View Delivery with ISL and DIBR (VDID)

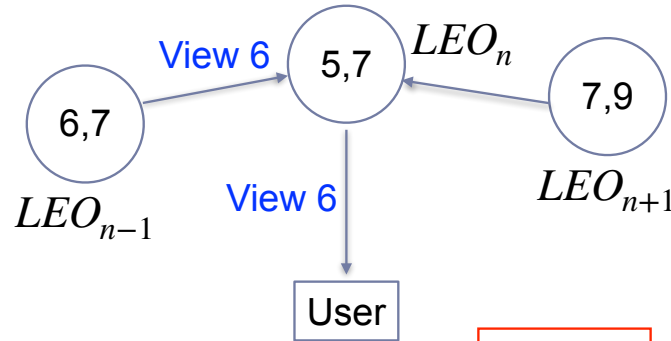
- ▶ User request $R_n(t) = 7, B = 5, D = 3 \rightarrow$ Request range: $[5,6,7,8,9]$

Step 1: $j = 5$ Local cache hit (5)



- ▶ $\tau_5 = c_{n,s}$
- ▶ $\mu_{5,5} = \tau_5$
- ▶ Transmit view 5 to user

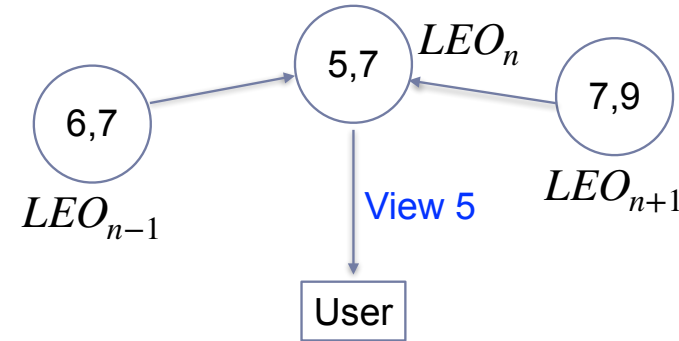
Step 2: $j = 6$ Neighbor cache hit (6)



- ▶ $\tau_6 = c_{ISL} + c_{n,s}$
- ▶ $\mu_{5,6} = \tau_6 + \mu_{5,5} = \tau_5 + \tau_6$
- ▶ Transmit view 5,6 to user

$$\mu_{5,5} = \tau_5$$

Step 3: $j = 7$ Local cache hit (7)



- ▶ $\tau_7 = c_{n,s}$
- ▶ $\mu_{5,7} = \tau_7 + \mu_{5,6}$
- ▶ $\tau_7 + \mu_{5,5} + [2\alpha + T_{DIBR}]$
- ▶ Transmit view 5,7 to user and synthesis view 6 on user

$$\mu_{5,6} = \tau_5 + \tau_6$$

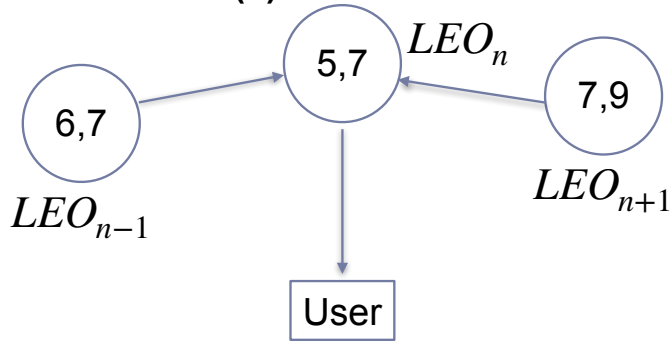


$$\mu_{h,j} = \min_{\max\{j-D, h\} \leq i < j} \{ \tau_j + \mu_{h,i} + [\alpha(j-i) + T_{DIBR}](j-i-1) \}$$

Phase 1: View Delivery with ISL and DIBR (VDID)

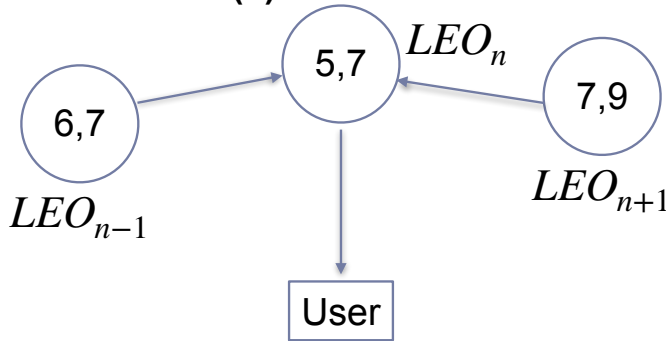
- ▶ User request $R_n(t) = 7, B = 5, D = 3 \rightarrow$ Request range: $[5,6,7,8,9]$

Step 4: $j = 8$
Cache miss (8)



- ▶ $\tau_8 = c_m + c_{n,f} + h \cdot c_{ISL} + c_{n,s}$
- ▶ $\mu_{5,8} = \tau_8 + \mu_{5,7}$
 $\tau_8 + \mu_{5,6} + [2\alpha + T_{DIBR}]$
 $\tau_8 + \mu_{5,5} + [3\alpha + T_{DIBR}] \cdot 2$
- ▶ Transmit view 5,8 to user and synthesis view 6,7 on user

Step 5: $j = 9$
Cache miss (8)



- ▶ $\tau_9 = c_{ISL} + c_{n,s}$
- ▶ $\mu_{5,9} = \tau_9 + \mu_{5,8}$
 $\tau_9 + \mu_{5,7} + [2\alpha + T_{DIBR}]$
 $\tau_9 + \mu_{5,6} + [3\alpha + T_{DIBR}] \cdot 2$
- ▶ Transmit view 5,6,9 to user and synthesis view 7,8 on user

Compute each view set

- ▶ $V_n^f(t) = \emptyset$
- ▶ $V_n^{DIBR}(t) = \{7,8\}$
- ▶ $V_n^{ISL}(t) = \{6,9\}$
- ▶ $V_n^s(t) = \{5,6,9\}$

$$\mu_{h,j} = \min_{\max\{j-D, h\} \leq i < j} \{ \tau_j + \mu_{h,i} + [\alpha(j-i) + T_{DIBR}](j-i-1) \}$$

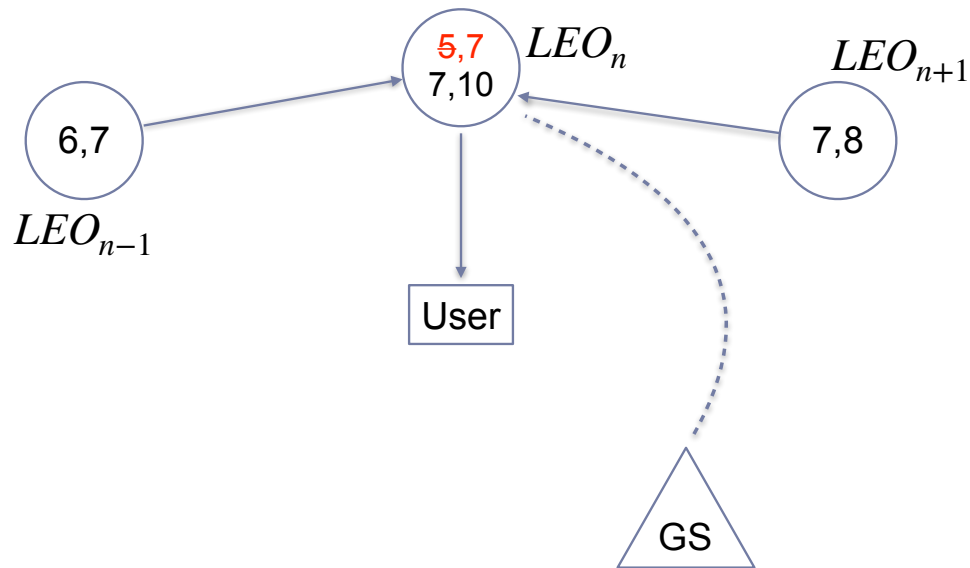
Phase 2: DIBR and Popularity-aware Cache Replacement (DPCR)

- ▶ Enhances the caching strategy by the infrequent connection between LEO and GS
 - ▶ Fetch popular view to avoid cache miss based on global popularity and contribution to synthesis
- ▶ Key Steps:
 1. Cache eviction
 - ▶ Remove views with lowest local popularity
 2. Capacity Management loop (until cache in size limit)
 - ▶ Global content fetching
 - ▶ Define an indicator function **popularity-synthesis score (PSS)** to get view importance
 - ▶ WPSS: $I_n(u) = P_{GS}(u) + \gamma \cdot \Delta_n^{fetch}(u)$
 - $P_{GS}(u)$: global popularity ranking from ground station
 - $\Delta_n^{fetch}(u)$: Number of additional views synthesized by adding u
 - ▶ Add optimal view to cache

Phase 2: DIBR and Popularity-aware Cache Replacement (DPCR)

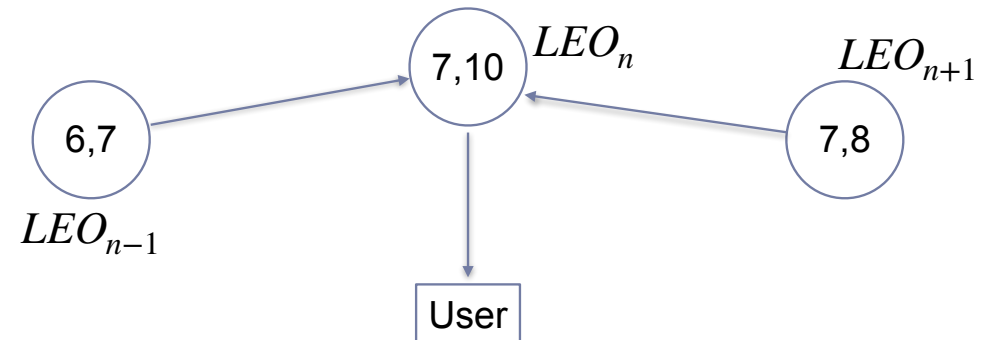
► $T = t :$

- LEO_n connect to GS
- LEO_n evict 5 and fetch 10



► $T = t + 1 :$

- $R_n(t + 1) = 8$
- Request range: [6,7,8,9,10]



System decide to transmit 6,8,10
and DIBR 7,9 at user

No cache miss is this
user request



Competitive Ratio and Time Complexity Analysis

► Competitive Ratio:

Theorem 2. *AMSCC* is $\min \left\{ \frac{3B}{p_n \cdot |V|}, \frac{3c_m D}{\underline{c}_{n,s}} \right\}$ – competitive online algorithm

- $p_n = p_n(|V|, \Delta_n, B, z) = 1 - \sum_{i=1}^{\Delta_n - (B-1)} f_n(i, z, |V|)$ the fraction of requests resulting in cache misses
- $f_n(i, z, |V|)$ is the Zipf distribution of the i -th most popular view [25]
- $\underline{c}_{n,s} = \min_{n,i} c_{n,s}(q_i, z_i, d_i)$ and Δ_n as the **maximum number of views** that can be hit in the cache

► Time Complexity:

- *AMSCC*: $O\left(T \cdot (|V| \log |V| + B\mathcal{Z}D) + |V| \mathcal{Z}^2\right)$
 - Set $\mathcal{Z} = \lfloor Z/z_i \rfloor$ is the maximum number of cached views in one cache

- [9] M. Yeh, C. -H. Wang, D. -N. Yang, J. -T. Lee and W. Liao, "Mobile Proxy Caching for Multi-View 3D Videos With Adaptive View Selection," in *IEEE Transactions on Mobile Computing*, vol. 21, no. 8, pp. 2909-2921, 1 Aug. 2022
- [19] T. E. Humphreys, P. A. Iannucci, Z. M. Komodromos and A. M. Graff, "Signal Structure of the Starlink Ku-Band Downlink," in *IEEE Transactions on Aerospace and Electronic Systems*, vol. 59, no. 5, pp. 6016-6030, Oct. 2023
- [20] A. U. Chaudhry and H. Yanikomeroglu, "Laser Intersatellite Links in a Starlink Constellation: A Classification and Analysis," in *IEEE Vehicular Technology Magazine*, vol. 16, no. 2, pp. 48-56, June 2021

Simulation Settings

- ▶ **Network Configuration:** [19,20]
 - ▶ 11 LEO satellites at 550km altitude
 - ▶ 25-degree minimum elevation angle
 - ▶ 2 ground stations
 - ▶ 100Gbps laser ISL, 2.5Gbps uplink
 - ▶ 250 MHz satellite-to-user bandwidth
- ▶ **Content & Caching:** [9,21,25,27]
 - ▶ 2500 multi-view videos, 16 views each, 32Mbps per view
 - ▶ 35GB per satellite, Zipf (0.8) popularity
 - ▶ Request range $B = 5$, DIBR synthesis distance $D = 3$
- ▶ **Algorithm Running setup**
 - ▶ Time slot length: 5 sec
 - ▶ The number of simulation time slots: 240
- ▶ **Users:** [26]
 - ▶ 500 users
 - ▶ Poisson arrival (0.25) and departure (0.15)
 - ▶ 15000 total requests over 240 time slots
- ▶ **Cost Parameters:** [9]
 - ▶ $c_m = 50, c_f = 20, c_{ISL} = 10$
 - ▶ $\alpha = 1, T_{DIBR} = 1$



- [21] Best video resolution for Android Devices | HeadJack. (2021, November 17). Headjack. <https://headjack.io/knowledge-base/best-video-resolution-for-android-devices/>
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- [26] Elizabeth Shriver, Arif Merchant, and John Wilkes. 1998. An analytic behavior model for disk drives with readahead caches and request reordering. SIGMETRICS Perform. Eval. Rev. 26, 1 (June 1998), 182–191.
- [27] J. -T. Lee, D. -N. Yang, Y. -C. Chen and W. Liao, "Efficient Multi-View 3D Video Multicast with Depth-Image-Based Rendering in LTE-Advanced Networks with Carrier Aggregation," in *IEEE Transactions on Mobile Computing*, vol. 17, no. 1, pp. 85-98, 1 Jan. 2018

Simulation Settings

▶ Baselines

- ▶ **EAYB** [9]: Mobile proxy caching adapted to LEO, DIBR synthesis, no ISL cooperation
- ▶ **SCA+EAYB** [22]: EAYB + satellite-based cooperative caching with K-means clustering
- ▶ **RFP+EAYB** [23]: EAYB + ridge regression prediction for regional content placement

▶ Experiments

- ▶ Variable:
 - Cache Size (% of total content size)
 - Number of views
 - Synthesis range D
- ▶ Performance metrics:
 - Hit rate
 - Total system cost
 - Average service latency
 - ISL hops

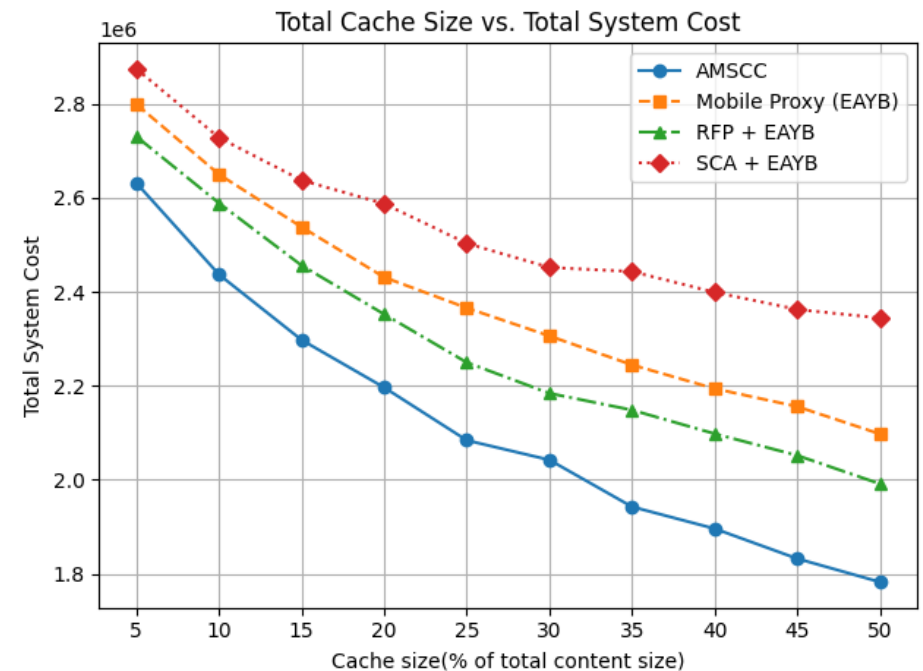
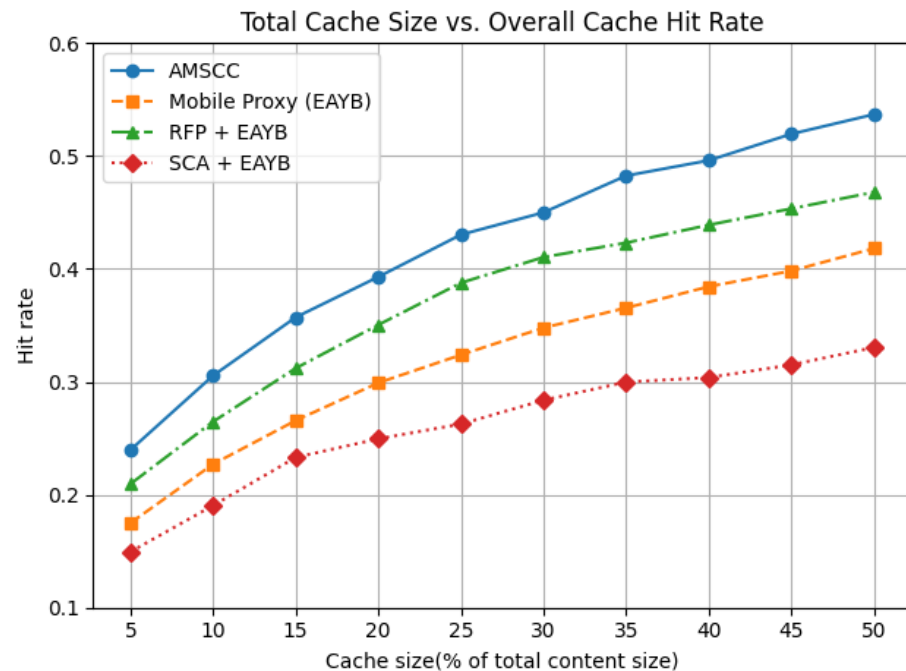
[9] M. Yeh, C. -H. Wang, D. -N. Yang, J. -T. Lee and W. Liao, "Mobile Proxy Caching for Multi-View 3D Videos With Adaptive View Selection," in *IEEE Transactions on Mobile Computing*, vol. 21, no. 8, pp. 2909-2921, 1 Aug. 2022

[22] S. Bhandari, T. X. Vu and S. Chatzinotas, "User-Centric Flexible Resource Management Framework for LEO Satellites With Fully Regenerative Payload," in *IEEE Journal on Selected Areas in Communications*, vol. 42, no. 5, pp. 1246-1261, May 2024

[23] J. Tang *et al.*, "Cooperative Caching in Satellite-Terrestrial Integrated Networks: A Region Features Aware Approach," in *IEEE Transactions on Vehicular Technology*, vol. 73, no. 7, pp. 10602-10616, July 2024

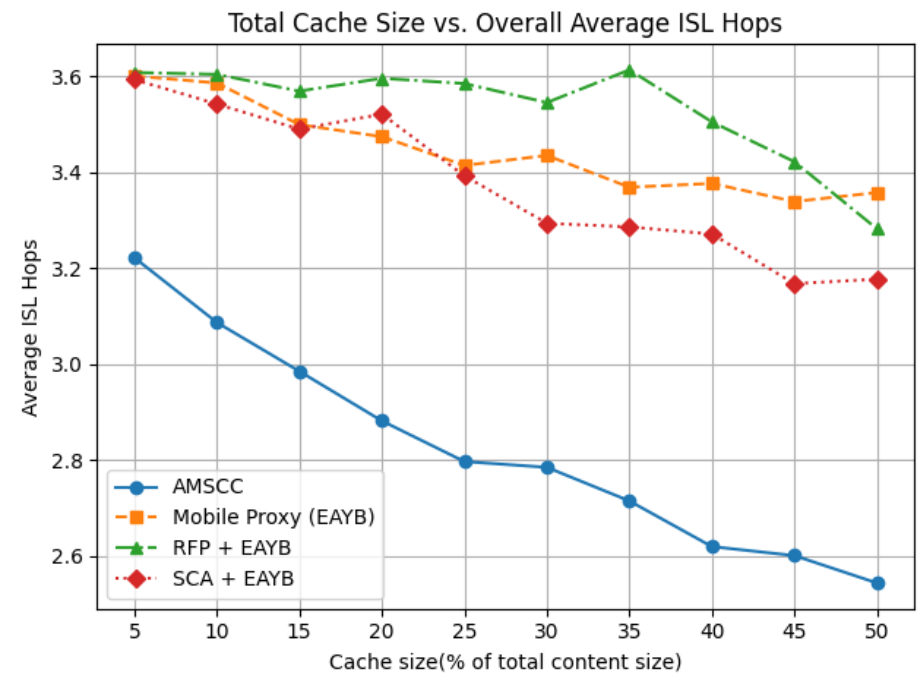
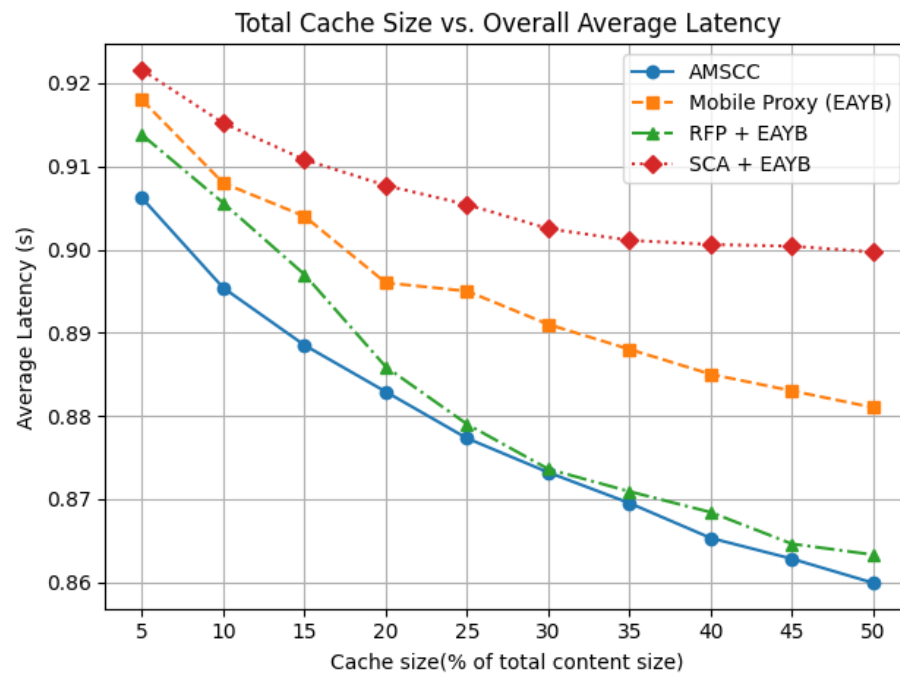
Simulation Results - Cache Size

- ▶ More storage enables better cooperative caching and synthesis strategies
- ▶ Strategic view placement exploits spatial correlations in multi-view videos
- ▶ Larger cache → Higher hit rates → Fewer expensive cache miss cost



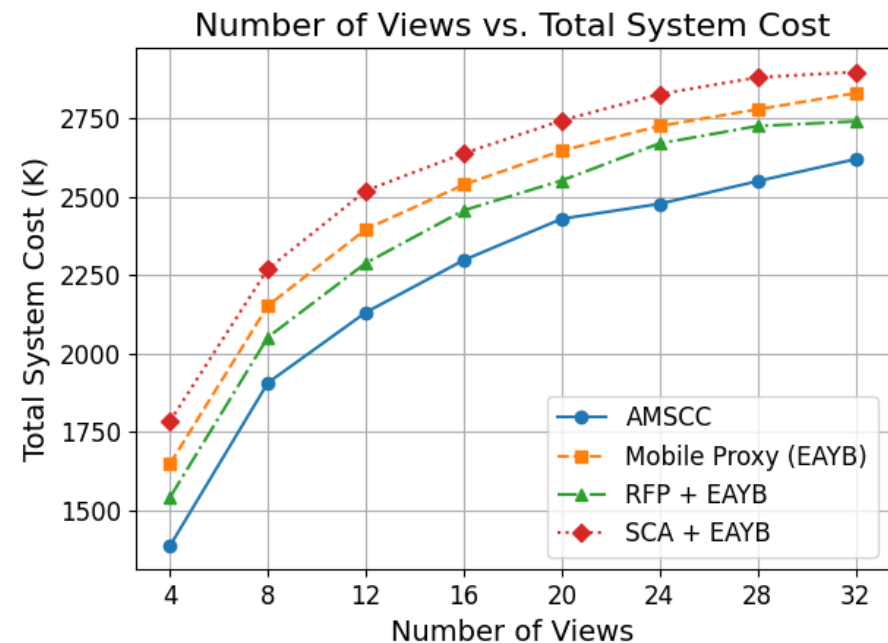
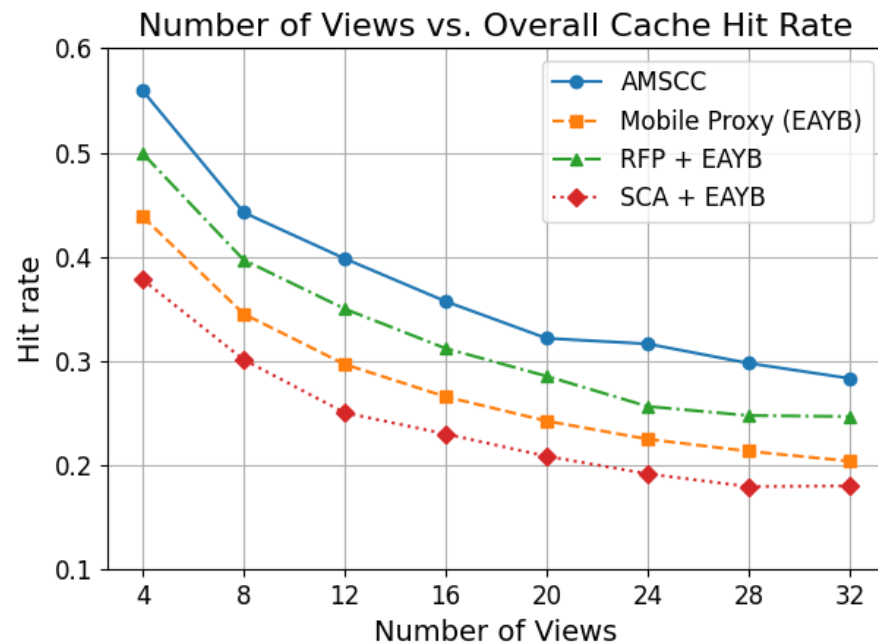
Simulation Results - Cache Size

- ▶ RFP+EAYB achieves best latency through advanced regional prediction
- ▶ AMSCC performs comparably through synthesis-aware placement, reducing cost
- ▶ AMSCC uses fewest ISL hops due to intelligent cooperative view positioning
- ▶ Larger cache → More local hits and coverage → Lower latency and ISL hops for all



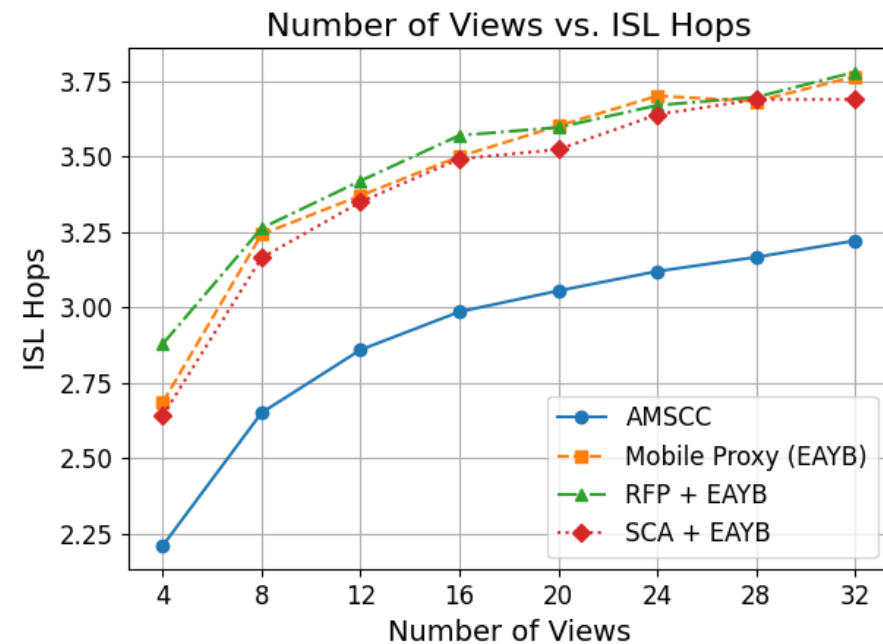
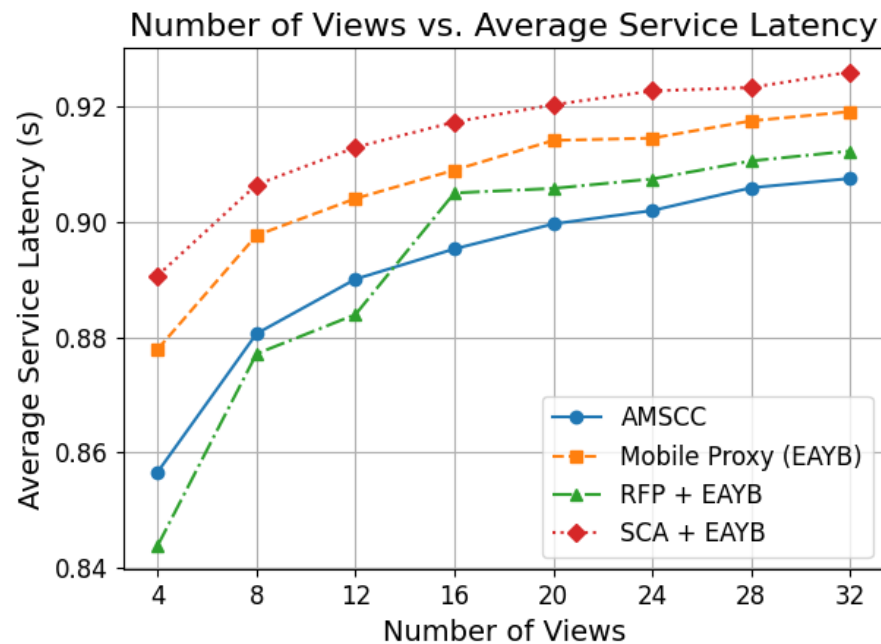
Simulation Results - Number of views

- ▶ AMSCC leverages DIBR synthesis to extend effective cache coverage
- ▶ Baselines treat views independently, missing spatial correlation benefits
- ▶ When the number of views is small, caching the most popular views of a video
 - generate a high hit rate, low total cost



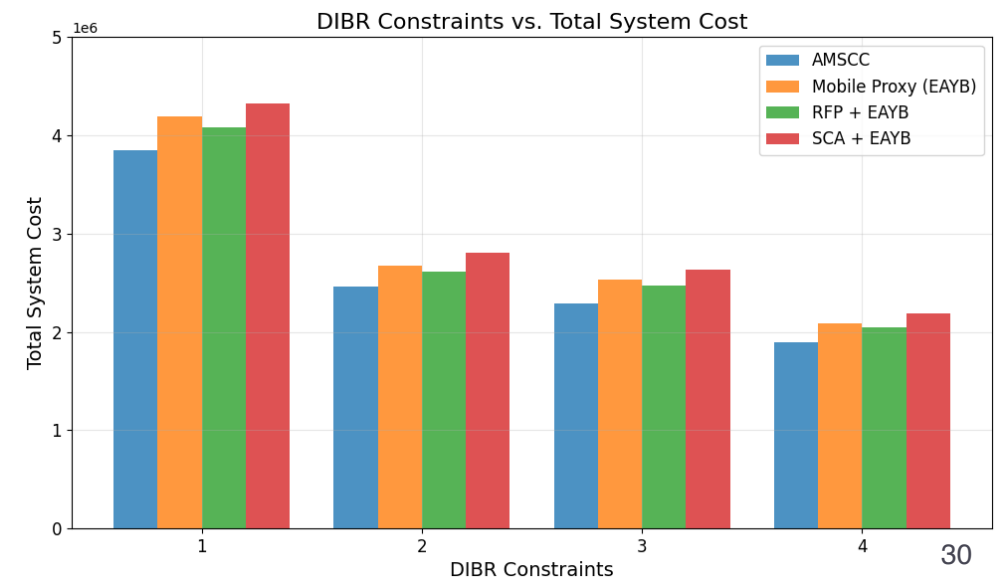
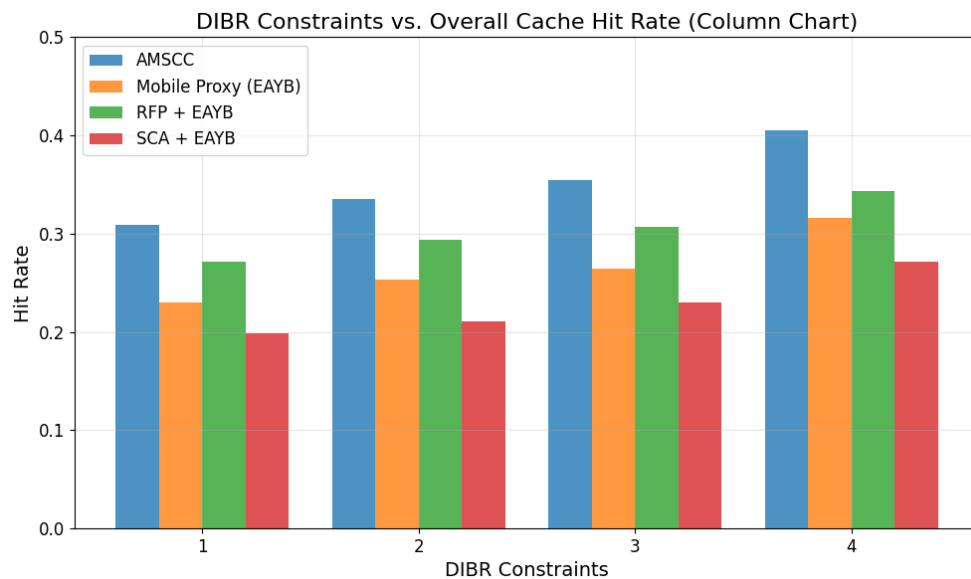
Simulation Results - Number of views

- ▶ AMSCC maintains competitive latency through efficient synthesis-aware caching and minimizes ISL communication overhead
- ▶ Baselines suffer from inefficient content placement as view diversity grows
- ▶ More views → Increased cache misses → Higher latency and hop counts



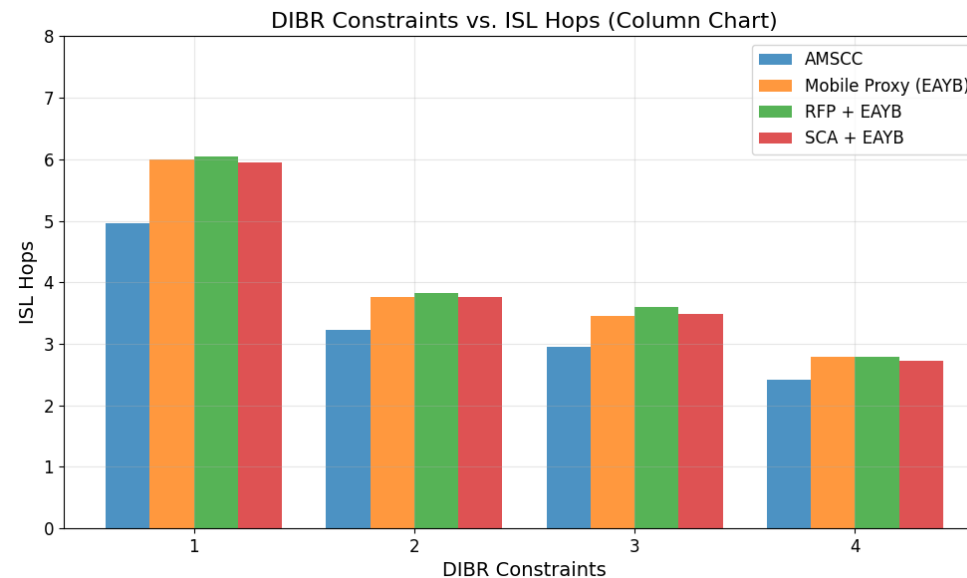
Simulation Results - Synthesis range D

- ▶ When the synthesis range D grows, the hit rate increases
 - More views can be acquired directly from the cache by DIBR
 - Larger D enables synthesis from more distant reference views
- ▶ $D=1$ to $D=2$ provides most significant cost savings, then gradually improves
- ▶ AMSCC considers DIBR synthesis across neighboring satellites (left, right, and self)



Simulation Results - Synthesis range D

- ▶ AMSCC cooperative view placement across neighboring satellites that minimizes ISL distance
- ▶ ISL Hops dramatically decrease as D increases



Conclusion

- ▶ Leverage ISL cooperation and DIBR synthesis to enhance cache effectiveness in LEO satellite networks for multi-view 3D video delivery
- ▶ New optimization problem MCCMS minimizing cache miss, fetching, user transmission, ISL and DIBR synthesis costs
- ▶ Proved MCCMS problem is NP-hard
- ▶ New algorithms: AMSCC with competitive ratio and time complexity analysis
- ▶ Achieve up to 35% higher cache hit rate and 10-30% cost reduction against baseline algorithms
- ▶ **Future Works:**
 - ▶ Investigate energy-aware cooperative caching for multi-view video in LEO satellite networks